HumidClimatologyEngine v11.5.0

Complete Scientific, Statistical, Engineering, Operational and Reproducibility Reference

Release identity: HumidClimatologyEngine 11.5.0

Schema: 11.5-FINAL | Checkpoint: 11.5-POWER-SAFE-BLOCKDAY

This document is intended to be a self-contained primary reference. A reader should not need a second technical reference to understand the data contract, processing model, statistics, failure handling, outputs, and interpretation.

Primary input contract: T2m + D2m + Surface Pressure (same primary input families used by the historical v8 workflow)

1. Executive overview

Chapter 1

v11.5 is the production successor of the hourly empirical v8 foundation. It preserves the same primary ERA5-Land input families while extending the software contract to three temporal resolutions, four decade products plus FULL 1981-2020, richer exact statistics, pair-wise dependence, thresholds, empirical joint structure, transactional recovery, audit and analysis.

Implementation contract

The central scientific principle is empirical-first: observed hourly states define the climatological sample. Parametric distributions are candidates, not hidden assumptions.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

2. Software identity and repository compatibility

Chapter 2

The software is version 11.5.0 even when the repository retains the historical filename `moisture_climatology_v8_0_FINAL_SINGLE_PASS_.py`. This is a compatibility naming decision, not a version claim.

Implementation contract

The schema and checkpoint versions are explicit in source and output metadata; v8 checkpoint files must never be mixed with v11.5 checkpoint files.

Key rules

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Chapter 3

v6 remains a historical/educational daily-statistical Gaussian/Monte-Carlo method. v7.5 established the hourly empirical production path and flexible modelling/reporting. v8 provided the direct hourly single-pass foundation. v11.5 turns that foundation into a richer multi-level and power-safe engine.

Implementation contract

Preserving old releases is necessary for scientific reproducibility; v11.5 does not rewrite their historical meaning.

Key rules

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4. Input contract

Chapter 4

Required inputs are t2m, d2m and sp. The default folders are the same three folders used by v8. File names are inventory aids; actual datetime coordinates are authoritative.

Implementation contract

Every month for every year must be uniquely identified. Missing or duplicate months are hard failures.

Key rules

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- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

5. Time axis and synchronization

Chapter 5

All three inputs must have compatible hourly timestamps. Pair statistics require synchronous observations; values from different timestamps must never be paired.

Implementation contract

The engine rejects duplicate/non-hourly axes and grid-coordinate mismatches rather than guessing.

Operational checklist

- Verify configuration fingerprint.
- Verify input inventory and grid.
- Verify sample support and masks.
- Retain logs and hashes.
- Do not infer unavailable quantities.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

6. Grid and spatial chunking

Chapter 6

The input grid is treated as canonical. Spatial chunks bound RAM and determine I/O granularity. Default chunk dimensions are 64 x 128.

Implementation contract

A chunk boundary is an operational boundary only; it must never change the scientific meaning of a cell.

Key rules

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- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

7. Climatological calendar

Chapter 7

The production calendar has 366 slots. Slots 1-58 map Jan-01..Feb-27, slot 59 is reserved, slot 60 pools Feb-28 and Feb-29, and slots 61-366 map Mar-01..Dec-31.

Implementation contract

The mapping is tested for leap and non-leap years and is part of the output schema.

Key rules

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- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

8. Physical variables

Chapter 8

The engine derives T, Td, P, saturation vapour pressure, vapor pressure e, relative humidity RH, mixing ratio r and specific humidity q.

Implementation contract

RH clipping is separated from physical validity; an invalid pressure partition is never made valid by clipping.

Key rules

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- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

9. Thermodynamic formulas

Chapter 9

Temperature is normalized to Celsius and pressure to hPa. Water/ice saturation formulas are selected by temperature sign.

Implementation contract

The implementation uses vectorized NumPy arithmetic with float32 working arrays and float64 statistical state.

Key rules

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- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

10. Validity masks and diagnostic counters

Chapter 10

Finite temperatures, finite positive pressure, positive vapor pressure and e<P are required for r/q. Supersaturation is counted separately.

Implementation contract

This distinction prevents invalid states from disappearing into a generic missing-value bucket.

Operational checklist

- Verify configuration fingerprint.
- Verify input inventory and grid.
- Verify sample support and masks.
- Retain logs and hashes.
- Do not infer unavailable quantities.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
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11. Temporal level model

Chapter 11

L1 is daily pooled. L2 has eight 3-hour bins. L3 has 24 hourly bins. The combined level dimension contains 33 bins.

Implementation contract

L1 is the primary daily climatology; L2 and L3 expose diurnal structure that would disappear under whole-day pooling.

Key rules

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- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

12. Scalar online statistics

Chapter 12

For RH/e/r/q the persisted state contains n, mean, M2, M3, M4, min, max, missing counts and threshold counts.

Implementation contract

Moments are accumulated online so raw multidecade samples need not be retained in RAM.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

13. Moment mathematics

Chapter 13

The engine uses mergeable central-moment state. Variance is M2/(n-1); higher moments are finalized only when their sample-size and positivity conditions hold.

Implementation contract

The merge equations are essential because decade states and FULL states must be combinable without replaying the raw archive.

Key rules

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- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

14. Extrema and threshold statistics

Chapter 14

Minimum and maximum are first-class outputs. RH thresholds, e thresholds, r thresholds and q thresholds are exact integer counters.

Implementation contract

Threshold counters are easier to audit than probabilities because the denominator and sample definition remain explicit.

Key rules

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- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

15. Bivariate paired state

Chapter 15

Configured pairs are RH-q, RH-e, q-e and r-q. Each pair retains paired-valid n, paired means, paired M2 values, Cxy, covariance and correlation.

Implementation contract

The pair-specific second moments are necessary so correlation is not accidentally computed from a different marginal mask.

Operational checklist

- Verify configuration fingerprint.
- Verify input inventory and grid.
- Verify sample support and masks.
- Retain logs and hashes.
- Do not infer unavailable quantities.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
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16. Empirical 2-D probability

Chapter 16

RH x q is the default empirical joint reference, represented as a piecewise-constant histogram. Default range is RH 0-130%, q 0-0.030 and resolution 8x8.

Implementation contract

Empirical joint structure is not a Gaussian surface and may retain skewness, multimodality and bounded behavior.

Key rules

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- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

17. Histogram design and limitations

Chapter 17

Histograms are persisted for L1 and L2 by default. L3 histograms are not stored by default to avoid disproportionate storage and sampling sparsity.

Implementation contract

Histogram binning is a resolution choice; sensitivity should be discussed in publication work.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

18. Quantile strategy

Chapter 18

When compressed quantile state is used, the representation is mergeable and compact rather than an exact raw-sample archive.

Implementation contract

A compressed quantile must be labelled approximate unless exact samples or exact order statistics are retained.

Key rules

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- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

19. Joint thresholds and event probabilities

Chapter 19

The engine tracks specified joint events such as RH>80 AND q>0.012. These counters can be normalized into probabilities or conditional frequencies.

Implementation contract

Joint event definitions are part of the configuration fingerprint and must be frozen for comparisons.

Key rules

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20. Four decade states

Chapter 20

v11.5 maintains 1981-1990, 1991-2000, 2001-2010 and 2011-2020 states.

Implementation contract

A year is routed to its decade and to FULL simultaneously, avoiding a second source replay solely to create FULL.

Operational checklist

- Verify configuration fingerprint.
- Verify input inventory and grid.
- Verify sample support and masks.
- Retain logs and hashes.
- Do not infer unavailable quantities.

Key rules

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21. FULL 1981-2020 and merge audit

Chapter 21

FULL is a direct all-year state. An independent audit can merge the four decade states and compare the result with FULL.

Implementation contract

A correct release requires the direct FULL and four-decade merge to agree within declared numerical tolerances.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

22. Transaction architecture

Chapter 22

The durable work unit is one calendar day x one spatial block. The journal records an explicit transaction identity and committed status.

Implementation contract

Progress telemetry is never allowed to define scientific completion.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

23. Before-image and crash recovery

Chapter 23

Before mutation, the existing state for the work unit is captured as a rollback image. The image is checksummed before the transaction becomes recoverable.

Implementation contract

If power is lost before commit, the incomplete unit is restored and replayed. If commit is durable, replay is forbidden.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

24. Why the old checkpoint failure cannot be repeated

Chapter 24

The historical failure involved checkpoint growth and an apparent advancement of processing state without a trustworthy relationship to the scientific state.

Implementation contract

v11.5 explicitly separates telemetry from truth and ties completion to a durable commit record for the same state update.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

25. Atomic metadata writes

Chapter 25

JSON metadata is written through temporary files, flushed, fsynced and replaced. Checkpoint NetCDF is synchronized before the commit marker is recorded.

Implementation contract

No application can promise durability beyond what the operating system and physical storage provide; the software nevertheless makes its transaction ordering explicit.

Operational checklist

- Verify configuration fingerprint.
- Verify input inventory and grid.
- Verify sample support and masks.
- Retain logs and hashes.
- Do not infer unavailable quantities.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

26. Performance engineering

Chapter 26

The objective is minimum wall-clock time subject to correctness, bounded memory and recoverability. Single-process execution eliminates worker synchronization and multiprocessing I/O contention.

Implementation contract

The remaining dominant costs at full scale are source-file access, compression and state persistence.

Key rules

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- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

27. Memory engineering

Chapter 27

Raw arrays are chunked. The active block is processed in vectorized form. State files for unrelated blocks are not kept open indefinitely.

Implementation contract

Memory tuning must be separated from scientific settings; changing chunk size must not change results.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

28. Finalization and NetCDF output

Chapter 28

Finalization is streamed by period, DOY and spatial block. Main statistics, diagnostics, bivariate parameters and histogram products are written with explicit metadata.

Implementation contract

The final outputs remain readable without loading the entire multidecadal state into memory.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

29. Output schema

Chapter 29

The primary dimensions are DOY, level-bin, latitude and longitude, with additional threshold and histogram dimensions as required.

Implementation contract

Metadata records engine version, schema, calendar, level contract, thresholds and histogram definitions.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
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30. Five-day local fitting layer

Chapter 30

The repository history contains a five-day centered-window distribution-fitting/query workflow. That workflow is an adjacent analytical layer and is not part of the primary accumulation hot path of the current v11.5 core file. The v11.5 core is authoritative for hourly accumulation, L1/L2/L3 statistics, extrema, thresholds, paired dependence state, empirical histogram, decade/FULL states, checkpoints, recovery, audit and final NetCDF products.

Implementation contract

The window supports local distribution fitting and station/grid-cell queries without redefining the primary climatology.

Operational checklist

- Verify configuration fingerprint.
- Verify input inventory and grid.
- Verify sample support and masks.
- Retain logs and hashes.
- Do not infer unavailable quantities.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

31. Candidate marginal distributions

Chapter 31

The historical fitting layer includes Normal, Skew-Normal, Pearson III, Beta for suitable bounded variables, and Bimodal Normal.

Implementation contract

Model selection is comparative; a candidate is not automatically scientifically correct.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

32. Copula and dependence modelling

Chapter 32

A Gaussian copula remains a reference candidate in the historical fitting layer. Additional family diagnostics can compare alternatives such as Student-t, Clayton, Gumbel and Frank.

Implementation contract

Copulas describe dependence, not marginal distributions; empirical RH-q remains the non-parametric reference.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

33. Model selection and dominance reporting

Chapter 33

Candidate fits can be ranked using AICc/BIC together with multimodality and tail diagnostics. A frozen model selection result can be converted into spatial and seasonal dominance maps.

Implementation contract

A dominance map shows where a candidate wins the declared scoring rule, not where a physical law has been proven.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

34. Quality assurance hierarchy

Chapter 34

Calendar, physics, moments, merge, pairs, histogram, recovery, serialization and analysis checks form a layered test pyramid.

Implementation contract

A production run is not accepted merely because it reached 100% progress.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

35. Provenance and reproducibility

Chapter 35

The run manifest should retain source identity, configuration fingerprint, script hash, input inventory, period, schema, checkpoint lineage and output hashes.

Implementation contract

The exact executed configuration is authoritative; README defaults are explanatory only.

Operational checklist

- Verify configuration fingerprint.
- Verify input inventory and grid.
- Verify sample support and masks.
- Retain logs and hashes.
- Do not infer unavailable quantities.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

36. Operational runbook

Chapter 36

Recommended sequence: selftest -> input validation -> small real-data pilot -> fault-injection recovery test -> production -> audit -> merge-audit -> analytical comparison -> archive.

Implementation contract

After interruption, retain checkpoints and restart with the same code/configuration.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

37. Analytical comparison framework

Chapter 37

The analysis layer must compare four decades and FULL across L1/L2/L3, variables, quantiles, extrema, thresholds, dependence, season and geography.

Implementation contract

Absolute change, percentage change, standardized change and threshold-risk change should be reported with sample support.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

38. Spatial and temporal science questions

Chapter 38

The engine can support questions about location, day-of-year, hour, decade, moisture variable, joint event and dependence changes.

Implementation contract

Spatial rankings must be accompanied by support and uncertainty indicators to avoid overinterpreting sparse cells.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

39. Publication and reporting rules

Chapter 39

Every figure/table should identify source period, variable, level, cell/region, sample count and software/schema version.

Implementation contract

Parametric fits must be labelled as models; empirical products must be labelled as data-driven references.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

40. Troubleshooting and failure classes

Chapter 40

Input, calendar, physics, state, transaction, output and analysis failures are distinct classes and should remain visible in logs.

Implementation contract

Do not bypass a checksum or manually mark a transaction as committed to make a run finish.

Operational checklist

- Verify configuration fingerprint.
- Verify input inventory and grid.
- Verify sample support and masks.
- Retain logs and hashes.
- Do not infer unavailable quantities.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

41. Migration and regression from v8

Chapter 41

Reuse the raw ERA5 archive, not v8 checkpoint states. Compare the common L1 RH/e/r/q products on controlled subsets first.

Implementation contract

New L2/L3, additional pair states, extrema and multi-period products are v11.5 features and require their own validation.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

42. Release acceptance criteria

Chapter 42

A release requires code selftest, input validation, real-data pilot, recovery test, merge audit, output schema check and provenance archive.

Implementation contract

The phrase '100% complete' should only refer to a verified release state, not merely a progress percentage.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

43. Appendix A - Configuration reference

Chapter 43

The scientific parameters should be frozen per release: period, chunking, level definitions, thresholds, histogram ranges, pairs and compression.

Implementation contract

Any scientific change requires a new schema/configuration fingerprint.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

44. Appendix B - Formula sheet

Chapter 44

All production formulas are listed in one place for audit and implementation review.

Implementation contract

The formula sheet should be used when independently reproducing calculations.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

45. Appendix C - Data and output schema

Chapter 45

The data schema defines dimensions, variable names, units, fill semantics and metadata.

Implementation contract

Downstream tools should inspect output metadata rather than infer undocumented defaults.

Operational checklist

- Verify configuration fingerprint.
- Verify input inventory and grid.
- Verify sample support and masks.
- Retain logs and hashes.
- Do not infer unavailable quantities.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

46. Appendix D - Recommended analytical report

Chapter 46

The report should contain executive summary, decadal tables, annual cycle, diurnal cycle, maps, extremes, dependence, QC and provenance.

Implementation contract

The same report should be regenerable from frozen output files without changing the science.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

47. Appendix E - Example research workflow

Chapter 47

Start with a few cells, inspect empirical behavior, validate statistics, then scale to the full grid.

Implementation contract

This staged workflow reduces the risk of discovering a conceptual issue after a multi-day production run.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

48. Final scientific position

Chapter 48

v11.5 is best understood as an empirical moisture-climatology and climate-distribution engine, not a forecast model or causal attribution system.

Implementation contract

Its strength is transparent observation-level processing, rich aggregation, restartability and auditable comparison across time and space.

Key rules

- Do not silently change the scientific definition while optimizing I/O.
- Treat COMMITTED state as the authoritative completion state.
- Keep version, schema and executed configuration visible in outputs.

Master release checklist

- Source file compiles in the target environment.
- Self-test passes.
- Input inventory is complete for the requested years.
- All three input grids match exactly.
- Calendar tests pass.
- Thermodynamic scalar reference tests pass.
- Online moments agree with trusted batch calculations.
- Merge equivalence passes.
- Paired covariance and correlation use pair-specific masks.
- Minimum and maximum are populated and sane.
- Threshold and joint-threshold counts are bounded by support.
- RH-q histogram count sum agrees with valid-pair support.
- Power-loss fault injection shows uninterrupted and recovered results agree.
- Four decade states exist.
- FULL state exists.
- FULL matches four-decade merge within declared tolerance.
- NetCDF outputs open and metadata are complete.
- Run manifest and hashes are archived.
- Analytical comparison report is reproducible.

Core-v11.5 versus analytical companion tools

A production repository is easier to trust when the primary calculation engine and expensive exploratory analysis are separated. The v11.5 core file is the authoritative accumulation engine. Companion scripts may consume its frozen outputs to perform five-day distribution fitting, alternative copula diagnostics, bivariate dominance visualization, and decadal comparison.

Authoritative core responsibilities

- Read and validate the three primary ERA5-Land input families.
- Derive RH, e, r and q.
- Accumulate L1/L2/L3 states.
- Persist exact moments, extrema, counts and thresholds.
- Persist pairwise dependence state.
- Persist the configured empirical RH-q histogram.
- Maintain four decade products and FULL_1981-2020.
- Provide transaction journal, before-image recovery and commit truth.
- Finalize auditable NetCDF products.

Companion analytical responsibilities

- Five-day centered local distribution fitting and station queries.
- Alternative marginal distribution comparison.
- Alternative copula-family diagnostics.
- Bivariate dominance maps.
- Decadal/spatial/seasonal comparison dashboards.
- Publication-ready PDF/HTML reports.

Why this separation matters

The primary climatology remains reproducible even when an analytical model family changes. A model-selection result can therefore be regenerated from frozen output files without recalculating 40 years of raw ERA5-Land input.